Credit Card Fraud Detection Using Cost-Sensitive Machine Learning

1. Introduction

With the rapid development of financial technologies and digital payment systems, the use of credit cards has significantly increased worldwide. While credit cards provide a fast and practical payment method for users, they also lead to an increase in financial fraud activities. Credit card fraud is an important security problem that can cause serious economic losses for banks, financial institutions, and users. Therefore, detecting fraudulent transactions as early as possible is of great importance in the financial sector.

The credit card fraud detection problem is a challenging classification problem from a machine learning perspective. The main reason for this difficulty is that fraudulent transactions represent only a very small proportion of the total transactions in real-world datasets. This leads to a highly imbalanced dataset structure. While fraudulent transactions constitute only a small percentage of the total transactions, normal transactions represent the vast majority of the dataset. For this reason, the accuracy metric, which is frequently used in classical classification problems, is not sufficient on its own to evaluate model performance. For example, a model that classifies all transactions as “normal” may achieve high accuracy but would not be useful in practice because it fails to detect fraudulent transactions.

In recent years, machine learning methods have been widely used in fraud detection problems. These methods can analyze large amounts of transaction data, learn patterns associated with fraudulent behavior, and generate automatic predictions for new transactions. In particular, algorithms such as Random Forest, Logistic Regression, Gradient Boosting, XGBoost, and LightGBM have shown strong performance in financial data analysis and anomaly detection problems.

However, increasing model accuracy alone is not sufficient in fraud detection problems. In real financial systems, the costs of different types of classification errors vary significantly. A fraudulent transaction that is not detected by the model (False Negative) can lead to direct financial losses for financial institutions. On the other hand, incorrectly labeling a normal transaction as fraud (False Positive) can negatively affect customer experience and cause unnecessary transaction blocks. Therefore, cost-sensitive approaches play an important role in fraud detection systems.

In this study, the credit card fraud detection problem is addressed using a cost-sensitive machine learning approach. First, a Random Forest model is trained as a baseline model and its performance is evaluated. Then, cost-based threshold optimization is applied by modifying the classification decision threshold. In this approach, the total cost is calculated as follows:

$$\text{Total Cost} = FP \times FP_cost + FN \times FN_cost$$

Where:

- FP (False Positive): A normal transaction incorrectly classified as fraud
- FN (False Negative): A fraudulent transaction missed by the model

In this study, the cost of missing a fraudulent transaction is defined as higher than the cost of a false alarm. Therefore, the FN cost is set higher than the FP cost. This approach aims to improve the model's ability to detect fraudulent transactions.

After performing threshold optimization for the Random Forest model, its performance is compared with other machine learning algorithms. In this context, Logistic Regression, Gradient Boosting, XGBoost, and LightGBM algorithms are trained and their performances are compared. The following metrics, which are more suitable for imbalanced datasets, are used for model evaluation:

- Precision
- Recall
- PR-AUC (Precision–Recall Area Under Curve)
- Cost-based evaluation

These metrics enable a more accurate evaluation of models in terms of fraud detection capability and false alarm rates.

In order to better interpret the model results, feature importance analysis is also conducted. This analysis identifies the variables that have the greatest impact on fraud detection. Additionally, SHAP (SHapley Additive exPlanations) is used to provide model interpretability and better understand the model’s decision-making mechanism.

To visually analyze and present the results more effectively, an interactive analysis dashboard is developed using Tableau. This dashboard visualizes model comparisons, the relationship between decision thresholds and cost, precision–recall analysis, and feature importance results.

Through the approach developed in this study, the fraud detection performances of different machine learning models are compared, and the most cost-effective model is identified. Furthermore, it is demonstrated that selecting the appropriate decision threshold has a significant impact on the performance of fraud detection systems.

2. Dataset Description

In this study, the publicly available Credit Card Fraud Detection dataset is used for the credit card fraud detection problem. The dataset contains credit card transactions made by European cardholders and represents real-world financial transaction data.

The dataset contains 284,807 transactions, of which 492 transactions are labeled as fraudulent. This indicates that the dataset is highly imbalanced. Fraudulent transactions account for approximately 0.172% of the dataset.

The dataset includes 31 variables. Among these variables, 30 are independent variables (features) and 1 is the target variable.

The target variable is named Class and represents a binary classification:

- Class = 0 → Normal transaction
- Class = 1 → Fraudulent transaction

Most of the variables in the dataset have been transformed and anonymized using Principal Component Analysis (PCA) for security and privacy reasons. Therefore, the features are labeled as V1, V2, V3, ..., V28. The PCA transformation preserves the structure required for modeling while protecting sensitive financial information.

There are two variables that have not been transformed by PCA:

- Time: Represents the time elapsed (in seconds) since the first transaction in the dataset.
- Amount: Represents the monetary value of the transaction.

```python
import numpy as np
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.metrics import (
     classification_report, confusion_matrix,
     average_precision_score, precision_recall_curve
)
from sklearn.ensemble import RandomForestClassifier
DATA_PATH = "creditcard.csv"
TARGET_COL = "Class"
TEST_SIZE = 0.30
RANDOM_STATE = 42
```

```python
df = pd.read_csv(DATA_PATH)
df.head()
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V21	V22	V23	V24	
0	0.0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599	0.098698	0.363787	...	-0.018307	0.277838	-0.110474	0.066928	0.1
1	0.0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803	0.085102	-0.255425	...	-0.225775	-0.638672	0.101288	-0.339846	0.
2	1.0	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461	0.247676	-1.514654	...	0.247998	0.771679	0.909412	-0.689281	-0.3
3	1.0	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609	0.377436	-1.387024	...	-0.108300	0.005274	-0.190321	-1.175575	0.6
4	2.0	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941	-0.270533	0.817739	...	-0.009431	0.798278	-0.137458	0.141267	-0.2

5 rows × 31 columns

The imbalanced structure of the dataset creates significant challenges for model training and evaluation. Therefore, stratified train-test split is used during dataset preparation.

This method ensures that the proportion of fraudulent transactions is preserved in both training and test datasets.

The dataset is divided as follows:

- 70% Training data
- 30% Test data

Due to the imbalanced nature of the dataset, accuracy alone is not sufficient for evaluating model performance. Therefore, Precision, Recall, and PR-AUC metrics are used. In addition, a cost-sensitive evaluation approach is applied considering that missing fraudulent transactions can lead to higher financial losses.

The dataset used in this study is one of the widely used benchmark datasets in fraud detection research and provides a suitable structure for comparing the performance of different machine learning algorithms.

3. Methodology

In this study, different machine learning algorithms are used for credit card fraud detection and the models are evaluated using a cost-sensitive approach. The methodology consists of several stages including data preparation, model training, decision threshold optimization, and model comparison.

3.1 Data Preparation

Before model training, the dataset is divided into training and test datasets. Considering the imbalanced structure of the dataset, stratified sampling is used.

This approach ensures that fraudulent transactions are distributed proportionally across both training and test datasets.

Dataset split:

- 70% Training data
- 30% Test data

The training dataset is used for model learning, while the test dataset is used to evaluate model performance.

3.2 Baseline Model: Random Forest

In this study, a Random Forest algorithm is used as the baseline model.

Random Forest is an ensemble learning method based on decision trees and is known for its strong performance on tabular data.

The model is trained using the following parameters:

- n_estimators = 300
- class_weight = balanced_subsample
- random_state = 42

The class_weight parameter is used to reduce the impact of class imbalance on the model.

After training, predictions are made on the test dataset and model performance is evaluated.

3.3 Model Evaluation Metrics

Since fraud detection involves highly imbalanced datasets, accuracy alone is not sufficient. Therefore, the following evaluation metrics are used:

- Precision: Indicates how many of the transactions predicted as fraud are actually fraudulent.
- Recall: Indicates how many of the actual fraudulent transactions are correctly detected by the model.
- PR-AUC (Precision–Recall Area Under Curve): Summarizes the relationship between precision and recall.

These metrics provide a more accurate evaluation for fraud detection problems.

3.4 Cost-Based Threshold Optimization

In fraud detection problems, the costs of different types of classification errors are not equal. Missing a fraudulent transaction (False Negative) can result in direct financial loss, while incorrectly labeling a normal transaction as fraud (False Positive) can negatively impact customer experience.

Therefore, a cost-sensitive threshold optimization approach is applied.

The total cost is calculated using the following formula:

$$\text{Total Cost} = FP \times FP_cost + FN \times FN_cost$$

In this study:

- FP cost = 10
- FN cost = 500

The model performance is evaluated for different threshold values and the optimal threshold that minimizes total cost is determined.

3.5 Model Comparison

After threshold optimization for the Random Forest model, its performance is compared with other machine learning algorithms:

- Logistic Regression
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM

Each model is trained on the training dataset and evaluated on the test dataset.

The models are compared based on:

- Precision
- Recall
- PR-AUC
- Total cost

This comparison allows analysis of each algorithm's ability to detect fraudulent transactions and its false alarm rate.

3.6 Feature Importance Analysis

To better understand the model's decision-making mechanism, feature importance analysis is performed.

This analysis identifies the variables that have the greatest impact on fraud detection.

Feature importance values are analyzed for both Random Forest and XGBoost models.

3.7 SHAP Model Explainability

To improve interpretability, the SHAP (SHapley Additive exPlanations) method is used.

SHAP calculates the contribution of each feature to model predictions.

The SHAP analysis examines:

- Global feature importance values
- Factors affecting model prediction for a single transaction

3.8 Data Visualization

The model results and analyses are visualized using Tableau.

The dashboard includes:

- Model comparison graphs
- Precision–Recall analysis
- Threshold–Cost relationship
- Feature importance visualizations

This dashboard enables a clear and interactive analysis of model performance and threshold optimization results.

4. Experimental Results (Results)

In this section, the performances of the machine learning models used in the study are compared and the results of the cost-sensitive approach are analyzed. The models are evaluated using Precision, Recall, PR-AUC, and total cost metrics.

4.1 Model Comparison

Within the scope of the study, fraud detection is performed using Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and LightGBM algorithms. Each model is evaluated on the test dataset, and cost-based threshold optimization is applied.

The obtained results are presented in the table below.

	model	best_cost	recall	precision	pr_auc
0	RandomForest	12080.0	0.844595	0.683060	0.821481
3	XGBoost	12560.0	0.858108	0.381381	0.808035
1	Logistic	13460.0	0.831081	0.561644	0.704996
2	GradientBoosting	13470.0	0.824324	0.721893	0.599682
4	LightGBM	27980.0	0.804054	0.081118	0.065563

Table 1. Performance comparison of machine learning models

When the table is examined, it can be observed that the Random Forest model achieved the lowest total cost. This indicates that the Random Forest model provides a balanced performance between successfully detecting fraudulent transactions and maintaining an acceptable false alarm rate.

Although the XGBoost model achieved a high recall value, it produced more false alarms due to its lower precision value. The Logistic Regression model, on the other hand, provided a more balanced performance; however, it resulted in a higher total cost compared to the Random Forest model.

The Gradient Boosting model demonstrated strong performance in terms of precision, but its overall performance remained limited due to a lower PR-AUC value. Meanwhile, the LightGBM model produced a very high recall value, but because its precision value was quite low, it became the weakest model in terms of cost.

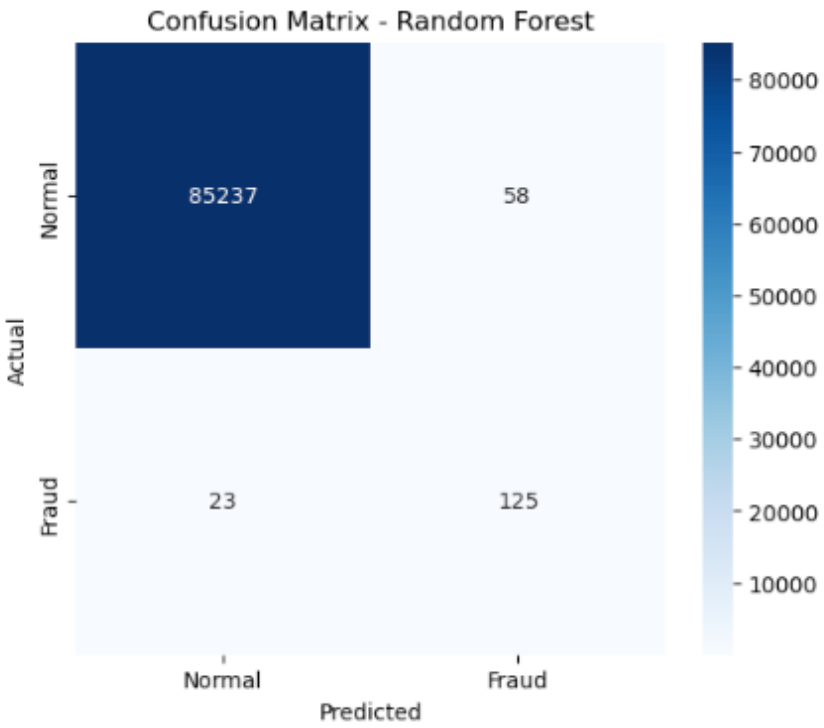

Figure 1. Random Forest model - confusion matrix

When the confusion matrix results are examined, it can be seen that the Random Forest model successfully detects a large proportion of fraudulent transactions. In addition, the model also correctly classifies the majority of normal transactions.

The obtained results indicate that the model achieves a high success rate in detecting fraudulent transactions while maintaining the false alarm rate at an acceptable level.

4.2 Decision Threshold and Cost Analysis

In fraud detection problems, the decision threshold of the model has a significant impact on performance. In this study, the total cost was calculated for different decision thresholds, and the threshold that produces the lowest cost value was determined.

Figure 2 shows the relationship between the decision threshold and the total cost for different models.

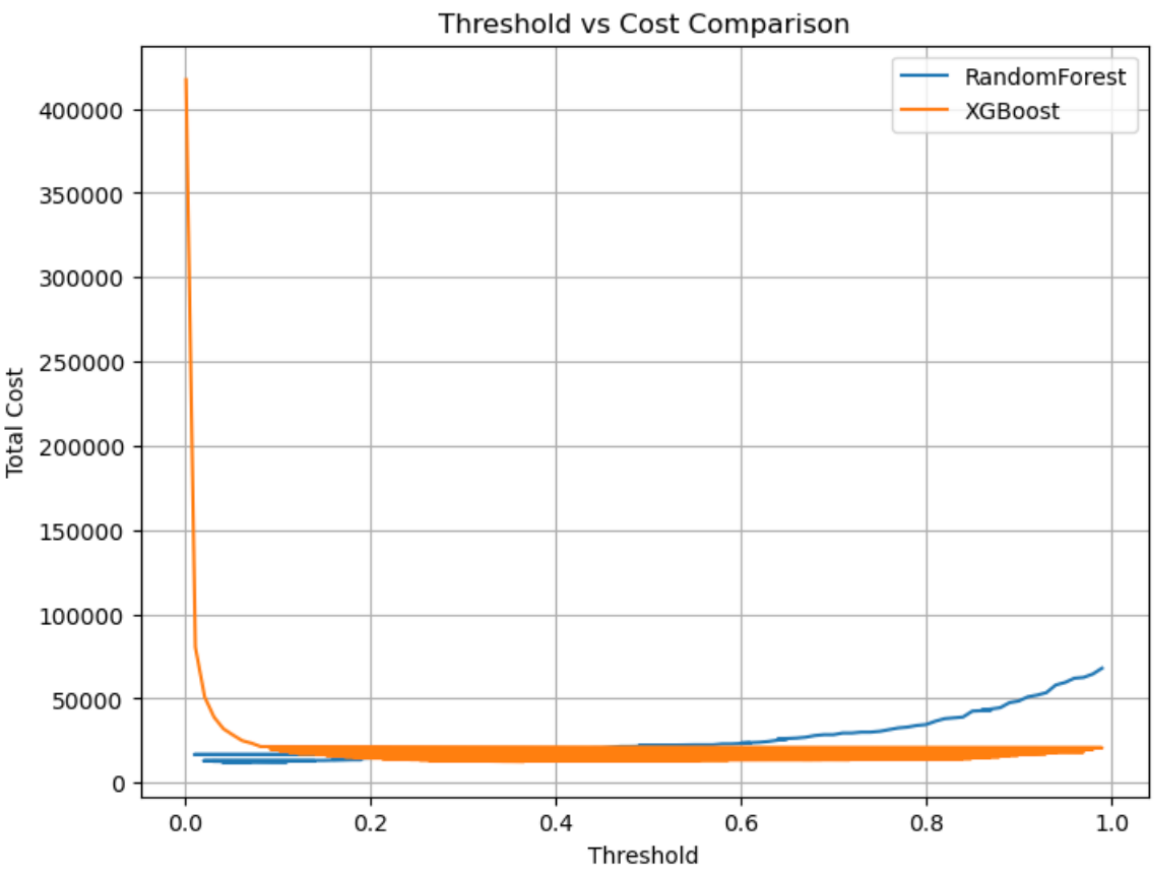

Figure 2. Relationship between decision threshold and total cost for Random Forest and XGBoost models

Figure 2 shows the total cost values obtained under different decision thresholds for the Random Forest and XGBoost models.

When the graph is examined, it can be seen that at low threshold values, the models predict fraud for too many transactions, which increases the number of false positives. This situation leads to an increase in the total cost. As the threshold value increases, the cost decreases up to a certain point; however, when the threshold becomes too high, the cost increases again due to missed fraudulent transactions. Therefore, determining the optimal decision threshold in terms of cost plays a critical role.

```python
best_t_xgb = float(best_xgb["threshold"])
pred_xgb_best = (proba_xgb >= best_t_xgb).astype(int)

print("XGB Best Threshold:", best_t_xgb)
print("XGB Best Cost:", int(best_xgb["cost"]))
print(confusion_matrix(y_test, pred_xgb_best))
print(classification_report(y_test, pred_xgb_best, digits=4))
```

```
XGB Best Threshold: 0.3643061224489796
XGB Best Cost: 12560
[[85089   206]
 [   21   127]]
              precision    recall  f1-score   support

           0     0.9998    0.9976    0.9987     85295
           1     0.3814    0.8581    0.5281       148

    accuracy                         0.9973     85443
   macro avg     0.6906    0.9278    0.7634     85443
weighted avg     0.9987    0.9973    0.9979     85443
```

Therefore, determining the correct decision threshold in fraud detection systems has a critical importance in terms of model performance.

4.3 Precision – Recall Analysis

Precision and Recall metrics are important indicators for evaluating model performance in fraud detection problems.

While Precision shows how many of the transactions predicted as fraud by the model are actually fraudulent, Recall shows how many of the actual fraudulent transactions are detected by the model.

Figure 3 compares the Precision and Recall values of the models.

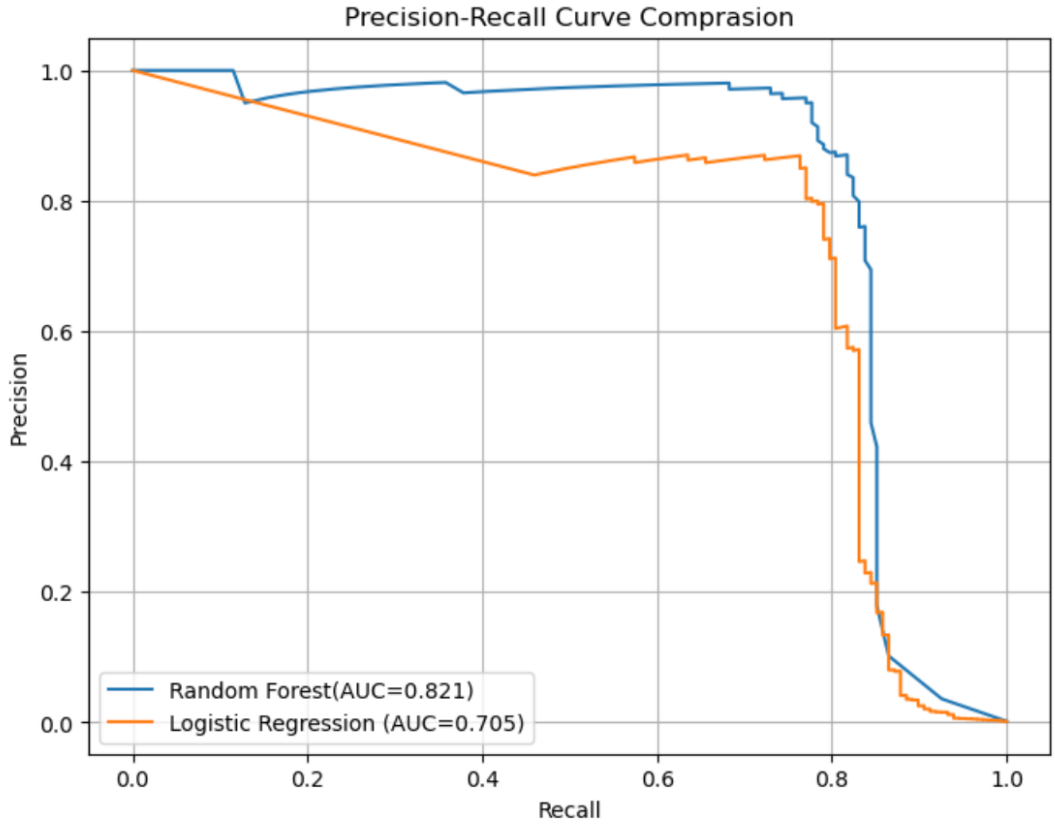

Figure 3. Precision–Recall curve for the Random Forest and Logistic Regression models

When the graph is examined, it can be seen that the Random Forest model provides a balanced performance between Precision and Recall. Although the XGBoost model has a higher Recall value, it produces more false alarms due to its lower Precision value.

In fraud detection systems, a high Recall value is generally preferred because missing fraudulent transactions can lead to financial losses. However, having a very low Precision value may also reduce the reliability of the system. Therefore, it is important to achieve a balanced performance between Precision and Recall.

4.4 Feature Importance Analysis

In order to better understand the decision-making mechanism of the model, a feature importance analysis was conducted. The contributions of variables to the model predictions were examined using Random Forest and XGBoost models.

As a result of the analysis, it was observed that the most influential variables in fraud detection are V4, V14, and V12. These variables stand out as the features that most strongly affect the model predictions.

Although it is not possible to directly interpret these variables due to the PCA transformation used in the dataset, it can be seen that these variables play an important role in distinguishing fraudulent transactions.

When the feature importance values calculated for the Random Forest model are examined, it is observed that the variables V14, V4, V10, and V12 play an important role in the model predictions.

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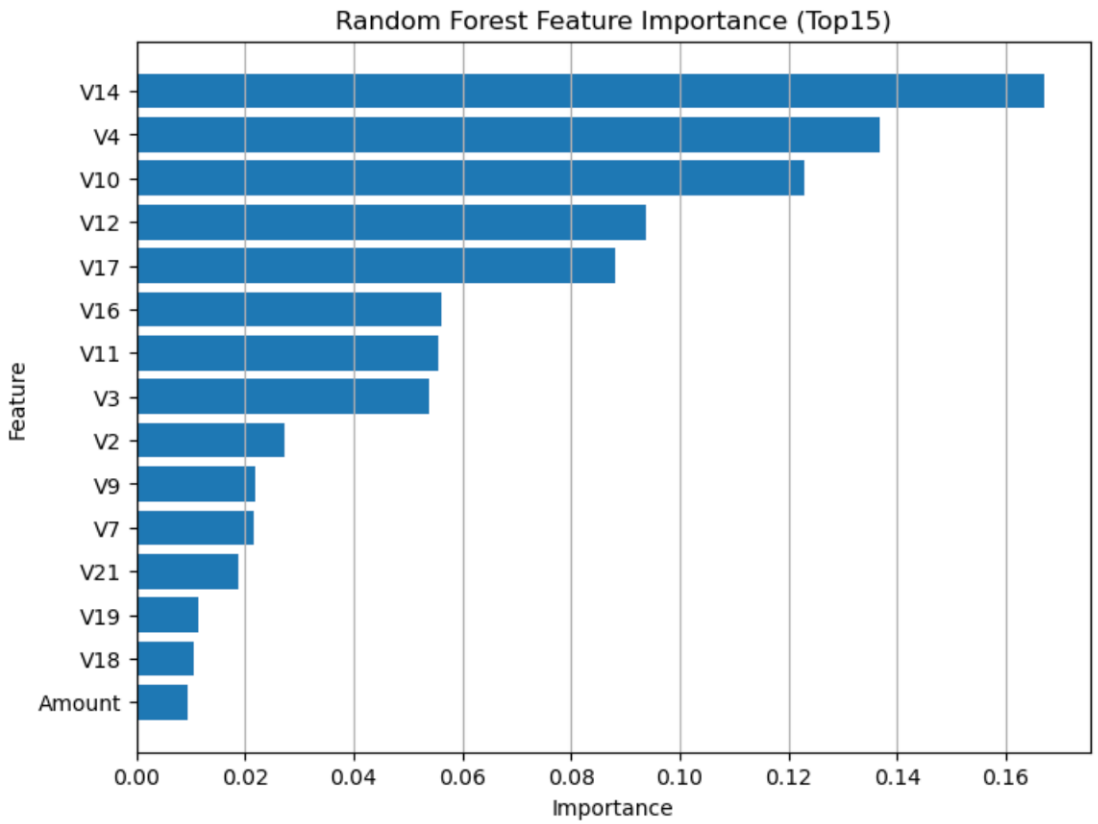

Figure 4. Feature importance graph for the Random Forest model

```
top_xgb = feat_imp_xgb.head(15).sort_values("importance", ascending=True)

plt.figure(figsize=(8,6))
plt.barh(top_xgb["feature"], top_xgb["importance"])
plt.xlabel("Importance")
plt.ylabel("Feature")
plt.title("XGBoost Feature Importance (Top 15)")
plt.grid(True, axis="x")
plt.show()
```

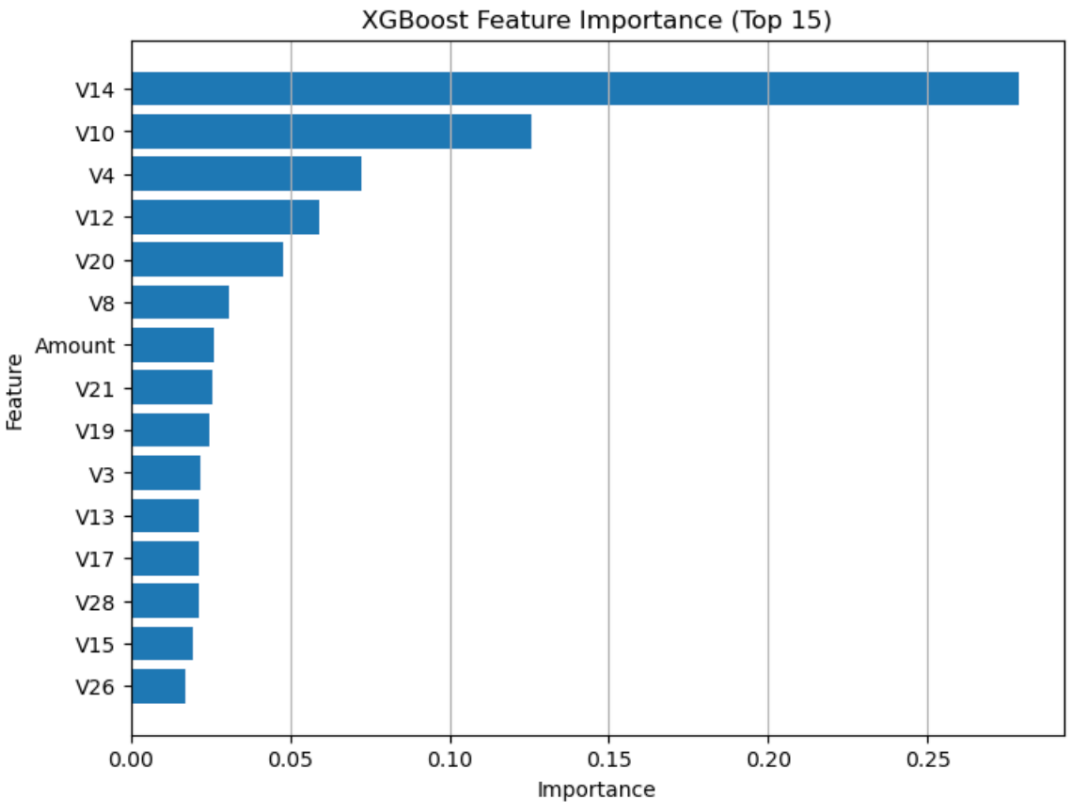

Figure 5 Feature importance graph for the XGBoost model

When Table 2 is examined, it can be seen that the most important variables for the Random Forest model are V14, V4, V10, and V12. These variables are the features that have the highest impact on the model predictions.

Although it is not possible to directly interpret the variables due to the PCA transformation used in the dataset, it can be said that these variables play an important role in distinguishing fraudulent transactions.

```
feat_imp = pd.DataFrame({
    "feature": X.columns,
    "importance": rf.feature_importances_,
    "model":"RandomForest"
}).sort_values("importance", ascending=False)
feat_imp.head(10)
```

	feature	importance	model
14	V14	0.167173	RandomForest
4	V4	0.136910	RandomForest
10	V10	0.123025	RandomForest
12	V12	0.093821	RandomForest
17	V17	0.088198	RandomForest
16	V16	0.056058	RandomForest
11	V11	0.055583	RandomForest
3	V3	0.053977	RandomForest
2	V2	0.027221	RandomForest
9	V9	0.021824	RandomForest

Table 2. Most important features according to the Random Forest model

4.5 SHAP Model Explainability

In order to better understand the model predictions, the SHAP method was used. Through SHAP analysis, the contributions of the variables affecting the model predictions were examined in detail.

The SHAP results support the global feature importance analysis and show that the variables V4, V14, and V12 play an important role in the model decisions.

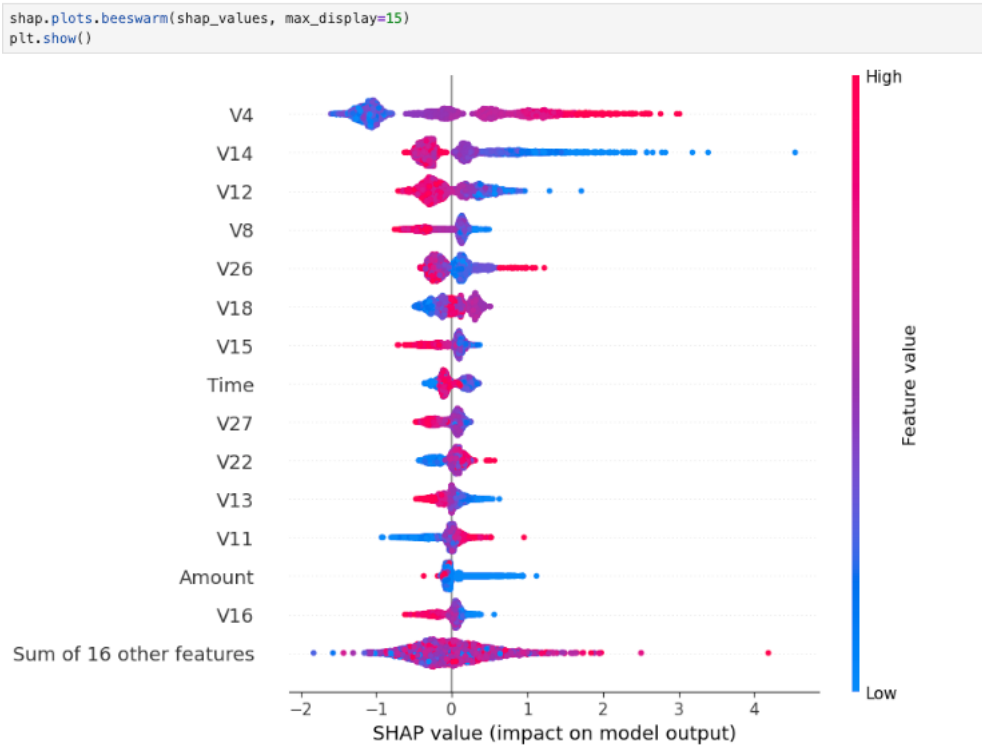

Figure 6. SHAP global feature graph

```
shap.plots.waterfall(shap_values[fraud_idx], max_display=15)
plt.show()
```

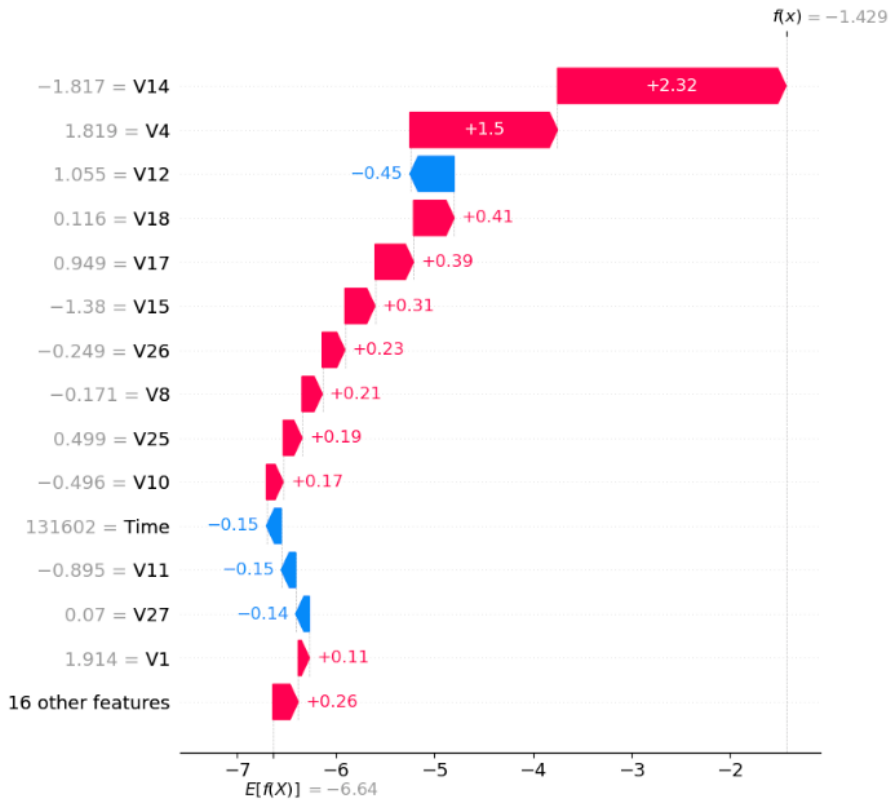

Figure 7.SHAP waterfall graph

```
shap.plots.bar(shap_values, max_display=15)
plt.show()
```

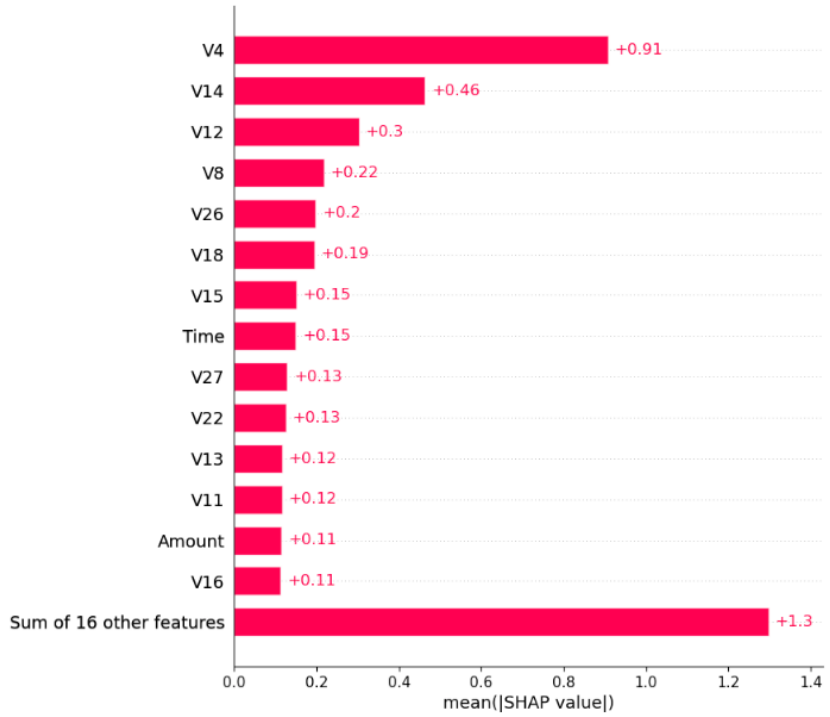

Figure 8. SHAP summary plot

In addition, by using SHAP waterfall graphs, it is visualized which variables affect the model prediction for a single transaction. Through this analysis, the model decisions become more transparent and interpretable.

4.6 Tableau Dashboard Analysis

In order to visually analyze the model results, an interactive dashboard was developed using Tableau.

The dashboard presents the following analyses:

- Model comparison graphs
- Precision – Recall analysis
- Threshold – Cost relationship
- Feature importance visualization

Through this dashboard, the performances of different models and the results of cost optimization can be easily analyzed.

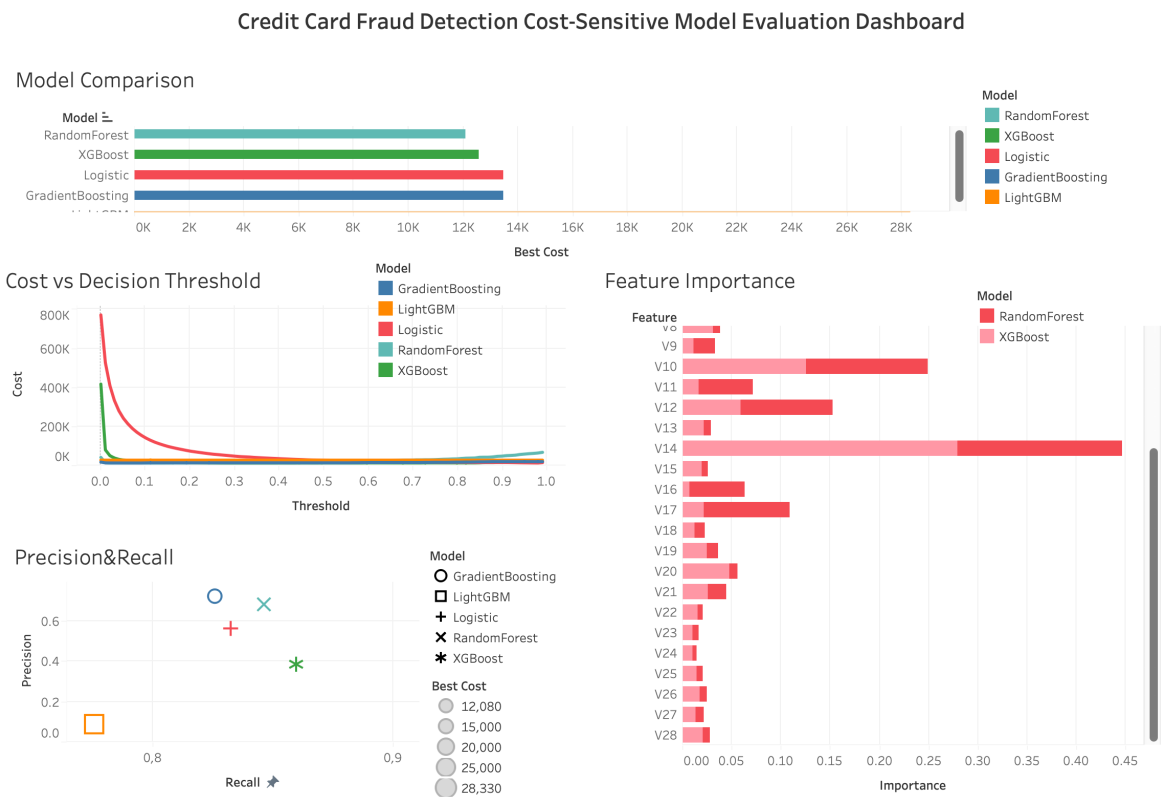

Figure 9. Fraud detection model comparison dashboard

5. Conclusion

In this study, the credit card fraud detection problem was addressed using a cost-sensitive machine learning approach. Different machine learning algorithms were trained on a highly imbalanced dataset consisting of financial transactions, and the performances of the models were compared using various evaluation metrics. In particular, precision, recall, PR-AUC, and total cost metrics, which are critical in fraud detection problems, were considered.

When the experimental results were examined, it was observed that the Random Forest model achieved the best performance in terms of total cost. The Random Forest model demonstrated a balanced performance in terms of both detecting fraudulent transactions and maintaining an acceptable false alarm rate. Although the XGBoost model produced high recall values, it generated more false positives due to its lower precision value. The Logistic Regression model, despite being a simpler model, provided a certain level of performance but remained behind the Random Forest model in terms of cost. Gradient Boosting and LightGBM models were also evaluated; however, they showed more limited performance, particularly in terms of precision and cost.

One of the important contributions of this study is the implementation of the cost-based threshold optimization approach. It is well known that False Negative errors in fraud detection systems can lead to significantly higher financial losses. Therefore, the classification decision threshold was adjusted in order to minimize the total cost. The obtained results show that determining the correct decision threshold has a significant impact on model performance. Through this approach, it becomes possible to detect fraudulent transactions more effectively.

To increase the interpretability of the model results, feature importance analysis and the SHAP (SHapley Additive exPlanations) method were used. SHAP analysis made model decisions more transparent and revealed that V4, V14, and V12 variables have significant effects on fraud detection. Such explainability methods are highly important for increasing the reliability of machine learning models used in financial systems.

The results obtained in the study were also visualized through an interactive dashboard developed using Tableau. Through this dashboard, the performances of different models, the relationship between the decision threshold and cost, and feature importance analyses can be visually examined. Such visualization tools enable machine learning results to be more easily understood not only by technical users but also by business decision-makers.

From a business perspective, fraud detection systems of this type provide significant operational advantages for banks and financial institutions. Machine learning-based fraud detection systems can reduce financial losses by identifying suspicious transactions at an early stage and improve risk management processes. In addition, the cost-sensitive modeling approach allows a more balanced structure between the false alarm rate and the fraud detection rate.

In future studies, model performance can be further improved by adding additional features to the dataset. Furthermore, deep learning-based methods, anomaly detection algorithms, or ensemble models can be used to enhance the performance of fraud detection systems. In addition, developing real-time fraud detection systems that operate on real-time transaction data stands out as an important research area.

In conclusion, this study demonstrates that cost-sensitive machine learning approaches can be effectively used in credit card fraud detection. The findings reveal that correct model selection and decision threshold optimization play a critical role in the performance of fraud detection systems.

6. Future Work

In this study, the credit card fraud detection problem was addressed using a cost-sensitive machine learning approach, and the performances of different machine learning models were compared. The results revealed that the Random Forest model achieved the best performance in terms of total cost. However, fraud detection is a continuously evolving research field, and model performance can be further improved through future studies.

First, model performance can be improved by applying advanced data preprocessing and feature engineering methods to the dataset. In particular, creating new features based on transaction behavior, analyzing user transaction habits, and incorporating time-series features into the model may contribute to detecting fraudulent transactions more accurately.

Second, in addition to the classical machine learning algorithms used in this study, deep learning-based methods can also be investigated. In particular, Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Autoencoder-based anomaly detection methods are among the powerful approaches that can be used for fraud detection in financial transaction data. These models may provide advantages in analyzing transaction sequences and user behaviors.

In addition, ensemble learning methods and model combinations can also be explored in fraud detection problems. By combining the outputs of multiple models, it is possible to obtain stronger and more reliable predictions. In particular, techniques such as stacking or blending can be useful for combining the strengths of different models.

Another important research area is the development of real-time fraud detection systems. For financial institutions, detecting fraudulent transactions as quickly as possible is critically important. Therefore, future studies may aim to optimize machine learning models so that they can operate on real-time transaction data.

Finally, model explainability and reliability are highly important in the financial sector. Therefore, future studies may examine model explainability methods more comprehensively and use different explainable artificial intelligence techniques to make model decisions more transparent. Such approaches can contribute to enabling financial institutions to use machine learning models more confidently in operational systems.

In conclusion, cost-sensitive machine learning approaches provide significant advantages in fraud detection problems. With future studies, more effective fraud detection systems can be developed by improving data analysis techniques, modeling methods, and real-time system integrations.